TED (21) – 2252 REVISION 2021

FIRST SEMESTER DIPLOMA EXAMINATION IN MANAGEMENT

BUSINESS MATHEMATICS

MODEL QUESTION PAPER –SET-1

Time: 3 hours Maximum Marks: 75

PART A

I. Answer all questions in one word or one sentence. Each question carries one mark.

(9 x 1 = 9 Marks)

1	A is a matrix of order p × q and B is a matrix of order q × r, then find the order of the product of matrix AB.	M 1.01	R
2	"Every identity matrix is a diagonal matrix". State whether the statement is True or False.	M 1.02	R
3	A die is thrown find the probability of getting a prime number.	M 2.02	U
4	Probability of the sure event is____________.	M 2.03	R
5	Calculate the Percentage if a student scores 90 marks out of total 120 marks.	M 3.03	U
6	After decreasing 24% in the price of an article, it costs Rs.912. Calculate the actual cost of the article.	M 3.04	A
7	Find the derivative of the function $f(x)=4x^2+5x^3+16$.	M 4.02	A
8	Find the Integral of $\int x^n\, dx$.	M 4.05	A
9	Calculate the total population of a village if 85% of the population of that village is 30600.	M 3.03	U

PART B

II. Answer any eight questions from the following. Each question carries 3 marks

(8 x 3 = 24 Marks)

1	Evaluate $\begin{vmatrix} 13 & -4 & 2 \\ 0 & 12 & 1 \\ 0 & 0 & 14 \end{vmatrix}$	M 1.02	A
2	If A=$\begin{bmatrix} 1 & 2 & 1 \\ 3 & 4 & 2 \\ 1 & 1 & 1 \end{bmatrix}$ and B=$\begin{bmatrix} 2 & 1 \\ 5 & 2 \\ 1 & 7 \end{bmatrix}$ verify that $(AB)^T = B^T A^T$	M1.03	A

3	A bag contains 10 red balls, 12 blue balls and 30 white balls. If a ball is drawn at random. What is the probability that is red.	M 2.02	A
4	Given P (A)=$\frac{1}{5}$ and P(B)=$\frac{3}{5}$. Find P(A or B), if A and B are mutually exclusive events.	M 2.03	A
5	Mr. Thomas invested an amount of Rs. 13,900 divided in two different schemes A and B at the simple interest rate of 14% p.a. and 11% p.a. respectively. If the total amount of simple interest earned in 2 years be Rs. 3508, calculate the amount invested in Scheme B.	M 3.01	A
6	A trader mixes 26 kg of rice at Rs. 20 per kg with 30 kg of rice of other variety at Rs. 36 per kg and sells the mixture at Rs. 30 per kg. Calculate his profit percentage.	M 3.04	A
7	In an election between two candidates, one got 55% of the total valid votes, 20% of the votes were invalid. If the total number of votes was 7500. Calculate the number of valid votes that the other candidate.	M 3.03	A
8	In a transaction, the profit percentage is 50% of the cost. If the cost further increases by 20% but the selling price remains the same, how much is the decrease in profit percentage.	M 3.04	A
9	Differentiate the following with respect to x (i) log $(sinx)$ (ii) $\frac{cosx}{\sqrt{x}}$	M 4.03	A
10	Enumerate the application of calculus in the field of business.	M 4.06	U

PART C

Answer all questions. Each question carries seven marks

(6 x 7 = 42 Marks)

III IV	Solve for x if $\begin{vmatrix} 1 & 2 & 0 \\ x & 3 & 1 \\ 0 & 5 & 4 \end{vmatrix} \neq 0.$ **OR** For the following matrices A and B, verify that (AB)' = B'A'. A $= \begin{bmatrix} 1 \\ -4 \\ 3 \end{bmatrix}$, $B = [-1 \quad 2 \quad 1]$.	M 1.01 M 1.03	A A
V	A die is rolled and a coin is tossed, find the probability that the die shows an odd number and the coin shows a head. **OR**	M2.02	A

VI	A card is drawn at random from a deck of cards. Find the probability of getting i) 8 of spade ii) a king iii) a queen of heart.	M2.02	A
VII	A sum of Rs 9600 is invested for 3 years at 10% p.a. at compound interest. 1. Calculate the sum due at the end of the first year. 2. Calculate the sum due at the end of the second year. 3. Find the compound interest earned in two years. 4. Find the difference between the answers in (2) and (1) and find the interest on this sum for one year. 5. Hence, write down the compound interest for the third year.	M3.02	A
	OR		
VIII	Enumerate the area of application of Profit & Loss calculation and Percentage calculation in various business scenarios.	M3.04	U
IX	A shopkeeper offers a discount of 20% on the selling price. On a special sale day, he offers an extra 25% off coupon after the first discount. If the article was sold for Rs. 3600, find a) The marked price of the article b) The cost price if the shopkeeper still makes a profit of 80% on the whole after all discounts are applied.	M3.03 M3.04	A
	OR		
X	A shopkeeper bought 600 oranges and 400 bananas. He found 15% of oranges and 8% of bananas were rotten. Find the percentage of fruits in good condition.	M3.03	A
XI	Find $\frac{dy}{dx}$, i) y= x^2 sec$x$ ii) y = $e^{\sec x + \tan x}$	M4.03	A
	OR		
XII	Find $\frac{dy}{dx}$, if y = $x^2 e^x sinx$	M4.03	A
XIII	Evaluate $\int x^2 sin\ x\ dx$.	M4.05	A
	OR		
XIV	Evaluate $\int \frac{sec^2 x}{cosec^2 x} dx$.	M4.05	A

SCHEME OF VALUATION
(Scoring Indicators)- Set I
BusinessMathematics

Qn. No	Scoring Indicators	Split up Score	Sub Total	Total
I	**PART A**			
1	The Order of Matrix AB is p * r	1	1	1
2	TRUE	1	1	1
3	The probability of getting a prime number is 1/2	1	1	1
4	The probability of a sure event is 1	1	1	1
5	Percentage is 75%	1	1	1
6	Actual Cost of article = 1200	1	1	1
7	Derivative is $8x+15x^2$	1	1	1
8	Integral is $\frac{X^{n+1}}{n+1}$	1	1	1
9	Total Population is 36000	1	1	1
II	**PART B**			
1	$\begin{vmatrix} 13 & -4 & 2 \\ 0 & 12 & 1 \\ 0 & 0 & 14 \end{vmatrix} = 13\begin{vmatrix} 12 & 1 \\ 0 & 14 \end{vmatrix}$ $= 13.14.15 = 2730$	1.5 1.5	3	3
2	$AB = \begin{bmatrix} 1 & 2 & 1 \\ 3 & 4 & 2 \\ 1 & 1 & 1 \end{bmatrix}\begin{bmatrix} 2 & 1 \\ 5 & 2 \\ 1 & 7 \end{bmatrix} = \begin{bmatrix} 13 & 12 \\ 28 & 25 \\ 8 & 10 \end{bmatrix}$ $(AB)^T = \begin{bmatrix} 13 & 28 & 8 \\ 12 & 25 & 10 \end{bmatrix}$	1.5	3	3

	$A^T = \begin{bmatrix} 1 & 3 & 1 \\ 3 & 4 & 1 \\ 1 & 2 & 1 \end{bmatrix}$ $B^T = \begin{bmatrix} 2 & 5 & 1 \\ 1 & 2 & 7 \end{bmatrix}$ $B^T A^T = \begin{bmatrix} 13 & 28 & 8 \\ 12 & 25 & 10 \end{bmatrix}$ $(AB)^T = B^T A^T$	1.5		
3	Total number of balls in the box, n(S)=10+12+30=52 Number of red balls, 10 Number of blue balls, 12 Number of white balls, 30 Hence the number of favorable outcomes for getting a red ball =10/52	1 1 1	3	3
4	P(A) =1/5 , P(B)= 3/5 Since A and B are mutually Exclusive, So P (AnB)=0 Now P(A or B) i.e P(AUB)= P(A)+ P(B)- P(A n B) 1/5 + 3/5 -0=4/5 Hence P(AUB)=4/5	1 1 1	3	3
5	Let the sum invested in Scheme A be Rs. x and that in Scheme B be Rs. (13900 - x). Then, (x × 14 × 2/100)+ [(13900 - x) × 11 x 2/100] = 2508 ⇒28x - 22x = 350800 - (13900 x 22) ⇒6x = 45000 ⇒x = 7500. So, sum invested in Scheme B = Rs. (13900 -7500) = Rs. 6400.	1 1 1	3	3
6	C.P. of 56 kg rice = Rs. (26 x 20 + 30 x 36) = Rs. (520 + 1080) = Rs. 1600. S.P. of 56 kg rice = Rs. (56 x 30) = Rs. 1680.	1	3	3

	$\text{Gair} = \left(\frac{(1680 - 1600}{1600} \times 100\right)\%$ $\Rightarrow \text{Gain} = \frac{80}{1500} \times 100\%$ $\Rightarrow \text{Gain} = 5\%$	1 1		
7	Total number of votes = 7500 Given that 20% of Percentage votes were invalid => Valid votes = 80% Total valid votes = 7500*(80/100) 1st candidate got 55% of the total valid votes. Hence the 2nd candidate should have got 45% of the total valid votes => Valid votes that 2nd candidate got = total valid votes x (45/100) 7500*(80/100)*(45/100) = 2700	1 1 1	3	3
8	Let us assume CP = Rs. 100. Then Profit = Rs. 50 and selling price = Rs. 150. The cost increases by 20% → New CP = Rs. 120, SP = Rs. 150. Profit % = 30/120 * 100 = 25%. Therefore, Profit decreases by 25%.	1.5 1.5	3	3
9	i. $\frac{d(\log(sinx))}{dx} = \frac{\frac{d(sinx)}{dx}}{sinx}$ $= \frac{cosx}{sinx} = cotx$ ii. $\frac{dy}{dx} = \frac{\sqrt{x}\frac{d}{dx}(cosx) - cosx\frac{d}{dx}(\sqrt{x})}{x}$ $= \frac{-(\sqrt{x}(sinx) + cos \quad \frac{1}{2\sqrt{x}}))}{x}$	1.5 1.5	3	3

10	1. Calculus is applies in finance and shares calculations 2. Calculus is applied in economics to understand marginal revenue and marginal cost. 3. In statistics calculus is used extensively Calculus is use in marketing research sales forecasting etc.	1*3	3	3
	PART C			
III	$\begin{vmatrix} 1 & 2 & 0 \\ x & 3 & 1 \\ 0 & 5 & 4 \end{vmatrix}$=1(3.4-5)-x(2.4-0) $\neq 0$ $\Rightarrow$ 7-8x$\neq$ 0 $\Rightarrow$ x$\neq \frac{7}{8}$	3 2 1 1	7	7
1V	$AB = \begin{vmatrix} -1 & 2 & 1 \\ 4 & -8 & -4 \\ -3 & 6 & 3 \end{vmatrix}$ $(AB)^T = \begin{vmatrix} -1 & 4 & -3 \\ 2 & -8 & 6 \\ 1 & -4 & 3 \end{vmatrix}$ $B^T A^T = \begin{vmatrix} -1 & 4 & -3 \\ 2 & -8 & 6 \\ 1 & -4 & 3 \end{vmatrix}$ $\therefore$ (AB)' = B'A'	2 2 2 1	7	7
V	Sample space S= {(1,H),(1,T),(2,H) ... (6,T)} Events E ={(1,H),(3,H),(5,H)} $P(E) = \frac{n(S)}{n(E)} = \frac{3}{12}$	2 2 3	7	7

VI	sample space S={ cards}, n(S)= 52 i. Event E ={8 of Spade}, n(E) = 1 Probability = $\frac{1}{52}$ ii. Event E ={a King}, n(E) = 4 Probability = $\frac{4}{52}$ iii. Event E ={a Queen of heart}, n(E) = 1 Probability = $\frac{1}{52}$	1 2 2 2	7	7
VII	It is given that Principal= 9600 Rate of interest = 10% p.a Period = 3 years We know that Interest for the first year= Prt/100 Substituting the values = (9600×10×1)/100 = 960 (i) Amount after one year = 9600 - 960= 10560 So the principal for the second year = 10560 Here the interest for the second year= (10560×10×1)/100 =1056 (ii) Amount after two years =10560+1056=11616 (iii) Compound interest earned in 2 years = 960+10560=2016 (iv) Difference between the answers in (ii) and (i) =11616−10560=1056 We know that Interest on 1056 for 1 year at the rate of 10% p.a.= (1056×10×1)/100 =105.60 (v) Here Principal for the third year = 11616 So the interest for the third year = (11616×10×1)/100	2	7	7

		5		
VIII	Application of Profit and loss calculations in Business a) In finance b) In Accounts c) In Marketing Application of Percentages calculations in Business a) In finance b) In Accounts c) In Marketing d) General Management	3 4	7	7
IX	Let the marked price of the article be x. First a 20% discount was offered, on which another 25% discount was offered. So, 75% of 80% of x = 3600 75/100 * 80/100 * x = 3600 → x = 6000. So the article was marked at Rs. 6000. Cost price of the article = [100/(100+80)]*3600 = Rs. 2000.	3 4	7	7
X	Total number of fruits shopkeeper bought = 600 + 400 = 1000 Number of rotten oranges = 15% of 600 = 15/100 × 600 = 9000/100 = 90 Number of rotten bananas = 8% of 400 = 8/100 × 400 = 3200/100 = 32 Therefore, total number of rotten fruits = 90 + 32 = 122 Therefore Number of fruits in good condition = 1000 - 122 = 878	2 2	7	7

	Therefore Percentage of fruits in good condition = (878/1000 × 100)% = (87800/1000)% = 87.8%	3		
XI	$i)\ \frac{dy}{dx} = x^2(secxtanx) + 2x(secx)$ $ii)\ y = e^{\sec x + \tan x}$ $logy = (secx + tanx)loge$ $\frac{1}{y}\frac{dy}{dx} = secxtanx + sec^2x$ $\frac{dy}{dx} = e^{secx+tanx}(secxtanx + sec^2x)$	3 4	7	7
XII	$\frac{d}{dx}(x^2e^xsinx)$ $= (x^2e^x)\frac{d}{dx}(sinx) + (x^2sinx)\frac{d}{dx}(e^x)$ $+ (e^xsinx)\frac{d}{dx}(x^2)$ $= x^2e^xcosx + x^2sinxe^x + 2xe^xsinx$	3 4	7	7
XIII	$\int x^2 sin\ x\ dx = x^2\int sinx\, dx - \int\left(\frac{d(x^2)}{dx}\int sin\, dx\right)dx$ $= -x^2cosx + \int(2xcosx)dx$ $= -x^2cosx$ $+ 2\left[x\int cosxdx - \int\left(\frac{d(x\ \ )}{dx}cosxdx\right)dx\right]$ $= -x^2cosx + 2xsinx - sinx$	2 3 2	7	7
XIV	$\int\frac{sec^2x}{cosec^2x}dx = \int\frac{\frac{1}{cos^2x}}{\frac{1}{sin^2x}}dx$ $\int tan^2xdx = \int(sec^2x - 1)dx$ $= \quad tanx - x + C$	2 3 2	7	7

TED (21) – 2252 REVISION 2021

FIRST SEMESTER DIPLOMA EXAMINATION IN MANAGEMENT

BUSINESS MATHEMATICS

MODEL QUESTION PAPER –SET-2

Time: 3 hours Maximum Marks: 75

PART A

I. Answer all questions in one word or one sentence. Each question carries one mark.

(9 x 1 = 9 Marks)

1	The ________________ of a matrix is a number that is specially defined only for square matrices.	M 1.01	R
2	If A = $\begin{bmatrix} 1 & 0 & 5 \\ 2 & 7 & 3 \\ 8 & 5 & 3 \end{bmatrix}$, find 9A.	M 1.02	U
3	Probability of getting a number multiple of 2 when a die is thrown is _________.	M 2.02	U
4	If the probability of winning a game is 0.4. Then find the probability of losing it.	M 2.01	R
5	Find the total population of the village if 96% of the population of the village is 23040.	M 3.03	A
6	Compound interest differs from simple interest in the fact that compound interest pays interests___________.	M 3.02	R
7	The derivative of a sum is the sum of the derivatives. State whether the statement is True or False.	M 4.01	R
8	Find the derivative of $x \log x$.	M 4.02	A
9	When at least one of the events occurs compulsorily from the list of events, then it is also known as____________.	M 2.03	R

PART B

II. Answer any eight questions from the following. Each question carries 3 marks

(8 x 3 = 24 Marks)

1	Explain different types of matrices with suitable examples.	M 1.02	U

2	If A = $\begin{bmatrix} 3 & -5 \\ -4 & 2 \end{bmatrix}$, show that $A^2 - 5A - 14I = 0$.	M1.01	A
3	A card is drawn at random from a deck of cards. Find the probability of getting, i) 3 of diamond ii) a queen	M 2.04	A
4	Three coins are tossed once. Find the probability of getting, i) atleast 3 heads ii) exactly 1 tail	M 2.04	A
5	A certain sum of money invested at the rate of 5% per annum compound interest, the interest compounded annually. If the difference between the interests in third year and first year is 102.50. Calculate the sum of money invested.	M 3.02	A
6	Father's weight is 25 % more than that of Son. Calculate by what percent is Son's weight less than Father's weight.	M 3.03	A
7	A fruit vendor bought 600 apples for Rs. 4800. He spent Rs. 400 on transportation. Calculate the selling price of each apple to get a total profit of Rs. 1000.	M 3.04	A
8	In a transaction, the profit percentage is 80% of the cost. If the cost further increases by 20% but the selling price remains the same, how much is the decrease in profit percentage.	M 3.03	A
9	Find $\int \frac{\sin(x-a)}{\sin(x+a)} dx$.	M 4.05	A
10	Find $\frac{□□}{□□}$ if $□ = \frac{□^2-1}{□^2+1}$.	M 4.03	A

PART C

Answer all questions. Each question carries seven marks

(6 x 7 = 42 Marks)

III	Find the value of *y-x* from the following equation. $2\begin{bmatrix} □ & 5 \\ 7 & □-3 \end{bmatrix} + \begin{bmatrix} 3 & -4 \\ 1 & 2 \end{bmatrix} = \begin{bmatrix} 7 & 6 \\ 15 & 14 \end{bmatrix}$	M1.01	A
	OR		
IV	Define Identity Matrix. If A=$\begin{bmatrix} \cos\alpha & -\sin\beta \\ \sin\beta & \cos\alpha \end{bmatrix}$ then for what value of α and β, A will be an Identity matrix.	M1.02	A
V	Two dies are thrown, what is the probability of following events, Getting the sum of numbers on the dies less than 3	M2.02	A

	Getting two numbers whose product is even Getting an even number in the first dice.		
	OR		
VI	A bag contains 100 red balls, 120 blue balls and 300 white balls. If a ball is drawn at random. What is the probability that it is, i) Blue ii) White.	M2.04	A
VII	A sum of Rs. 50,000 is borrowed and the rate of interest is 10% per annum. What is the compound interest for 5 years.	M3.02	A
	OR		
VIII	Differentiate between Simple interest and Compound interest.	M3.01 M3.02	U
IX	A dishonest merchant sells his grocery using weights 15% less than the true weights and makes a profit of 20%. Find his total gain percentage.	M3.03, M3.04	A
	OR		
X	A shopkeeper bought 1600 oranges and 1400 bananas. He found 15% of oranges and 8% of bananas were rotten. Find the percentage of fruits in good condition.	M3.03	A
XI	If □ = □□□□^{3}□, □ = □□□□^{3}□, find the value of $\frac{□□}{□□}$.	M 4.03	A
	OR		
XII	If = □$^{□□□□-□□□□}$, find $\frac{□□}{□□}$.	M4.03	A
XIII	Find $\int (\log x)^2$ □ □□.	M4.05	A
	OR		
XIV	Evaluate$\int$ □^{2}□$^{2□}$□□.	M4.05	A

SCHEME OF VALUATION
(Scoring Indicators)- Set II
BusinessMathematics

Qn. No	Scoring Indicators	Split up Score	Sub Total	Total
I	**PART A**			
1	Determinants	1	1	1
2	$\begin{bmatrix} 9 & 0 & 45 \\ 18 & 63 & 27 \\ 72 & 45 & 27 \end{bmatrix}$	1	1	1
3	Probability of getting a number multiple of 2 when a die is thrown is 1/2	1	1	1
4	The probability of losing the event is 0.6	1	1	1
5	Total Population is 24000	1	1	1
6	On interest as well	1	1	1
7	TRUE	1	1	1
8	1+ logX	1	1	1
9	Exhaustive	1	1	1
II	**PART B**			
1	Row Matrix, Column Matrix, Scalar Matrix, Square Matrix, Identity Matrix, Rectangular Matrix. Explain any 3 with example	1 1 1	3	3
2	A = $\begin{bmatrix} 3 & -5 \\ -4 & 2 \end{bmatrix}$, $\square^2 = \begin{bmatrix} 29 & -25 \\ -20 & 24 \end{bmatrix}$, 5 A = $\begin{bmatrix} 15 & -25 \\ -20 & 10 \end{bmatrix}$,	1	3	3

	$14\square = \begin{bmatrix} 14 & 0 \\ 0 & 14 \end{bmatrix}$ $\square^2 - 5\square - 14\square = 0$	1 1		
3	sample space S={ cards}, n(S)= 52 i. Event E ={3 of diamond}, n(E) = 1 Probability = $\frac{1}{52}$ ii. Event E ={a queen}, n(E) = 4 Probability = $\frac{4}{52}$	1 1 1	3	3
4	Sample space S={ HHH,HHT,HTT,HTH,THH,THT,TTH,TTT}, n(S)= 8 i. Event E ={HHH}, n(E) = 1 Probability = $\frac{1}{8}$ ii. Event E ={HHT,THH,HTH}, n(E) = 3 Probability = $\frac{3}{8}$	1 1 1	3	3
5	Let Principal = P → Rate = 5% per annum compounded Annually . → Diff. b/w CI of 3rd year - CI of first year = **Rs.102.5** **<u>First Year</u> :-** → Amount = P(1+R/100)^T → A = P(1 + 5/100)^1 → A = P(1 + 0.05) → A = 1.05P **So,** → CI of first year = Amount - Principal = 1.05P - P = **0.05P.** **<u>Similarly Second Year</u> :-**	1 2	3	3

	→ CI till 2nd year = Amount - Principal = 1.1025P - P = **0.1025P**. **<u>Similarly Third Year</u> :-** → CI till 3rd year = Amount - Principal = 1.157625P - P = 0.157625P **Now,** → CI for third year = CI till third year-CI till second year → CI for third year = 0.157625P - 0.1025P = **0.055125P** **Therefore,** → Diff. b/w 3rd year CI & 1st year CI = 0.055125P - 0.05P = **0.005125P.** → 0.005125P = 102.5 → P = (102.5) ÷ (0.005125) → P = **Rs.20,000 (Ans.)**			
6	Let son's weight be 100 kg. Then father's weight = (100 + 25) kg = 125 kg If father's weight is 125 kg, then son's weight is 100 kg. If father's weight is 1 kg, then son's weight is 100/125 kg If father's weight is 100 kg, then son's weight = (100/125 × 100) kg Therefore, son's weight is 20 % less than that of father.	1 1 1	3	3
7	Cost of 600 apples = 4800 Transportation = 400 Total cost= 5200 Profit required= 1000 Total SP= 6200 Sp of 1 apple= 6200/600=**10.33**	1 1 1	3	3

8	Let us assume CP = Rs. 100. Then Profit = Rs. 80 and selling price = Rs. 180. The cost increases by 20% → New CP = Rs. 120, SP = Rs. 180. Profit % = 60/120 * 100 = 50%. Therefore, Profit decreases by 30%.	1 2	3	3
9	$\int \frac{\sin(x-a)}{\sin(x+a)} \square\square = \int \frac{\sin(t-a-a)}{\sin t} \square\square = \int \frac{\sin(t-2a)}{\sin t} \square\square$ $= \int \frac{\sin t.\cos 2a - \cos t.\sin 2a}{\sin\ t} \square\square$ $= \int (\square\square\square 2\square - \square\square\square\square.\square\square\square 2\square)\ \square\square$ $= \square\square\square 2\square\ .\ \square - \square\square\square 2\square.\log(\lvert\square\square\square\square\rvert)$ $= \square\square\square 2\square\ .\ (\square + \square) - \square\square\square 2\square.\log(\lvert\sin(\square + \square)\rvert) + \square$	1 1 1	3	3
10	$\frac{\square\square}{\square\square} = \frac{\square}{\square\square}\left(\frac{\square^2 - 1}{\square^2 + 1}\right)$ $\square = \frac{(\square^2 - 1) + 1 - 1}{\square^2 + 1} = \square = \frac{(\square^2 + 1) - 2}{\square^2 + 1}$ $\square = 1 - \frac{2}{\square^2 + 1}$ $\frac{\square\square}{\square\square} = \frac{\square}{\square\square}\left(1 - \frac{2}{\square^2 + 1}\right)$	1 1 1	3	3

	$= \frac{4\square}{(\square^{2}+1)^{2}}$			
	PART C			
III	$2\begin{bmatrix} \square & 5 \\ 7 & \square - 3 \end{bmatrix} + \begin{bmatrix} 3 & -4 \\ 1 & 2 \end{bmatrix} = \begin{bmatrix} 7 & 6 \\ 15 & 14 \end{bmatrix}$ $\begin{bmatrix} 2\square & 10 \\ 14 & 2\square - 6 \end{bmatrix} + \begin{bmatrix} 3 & -4 \\ 1 & 2 \end{bmatrix} = \begin{bmatrix} 7 & 6 \\ 15 & 14 \end{bmatrix}$ $\begin{bmatrix} 2\square + 3 & 6 \\ 15 & 2\square - 4 \end{bmatrix} = \begin{bmatrix} 7 & 6 \\ 15 & 14 \end{bmatrix}$ $2\square + 3 = 7,\ \ 2\square - 4 = 14$ $\square = 2, \square = 9.$ $\square - \square = 7$	2 2 2 1	7	7
1V	Definition of Identity Matrix A=$\begin{bmatrix} \square\square\square\square & -\square\square\square\square \\ \square\square\square\square & \square\square\square\square \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ $\square\square\square\square = 1,\ \ \square\square\square\square = 0$ $\square = 2\square\square,\ \ \square \in \square$ $\square = \square\square,\ \ \square \in \square$	1 3 3	7	7
V	sample space S={ (1,1),(1,2), ... , (6,6)}, n(S)= 36 i. Event E ={(1,1)}, n(E) = 1 Probability = $\frac{1}{36}$ ii. Event E ={(1,2),(1,4),(1,6), ... , (6,6)}, n(E) = 24	1 2 2	7	7

	Probability = $\frac{24}{36}$ iii. Event E ={(2,1),(2,2),(2,3), ... , (6,6)}, n(E) = 18 Probability = $\frac{18}{36}$	2		
VI	Total number of balls in the box, n(S)=100+120+300=520	1	7	7
	Number of red balls, 100 Number of blue balls, 120	2		
	Number of white balls, 300 Hence the number of favorable outcomes for getting a	2		
	Blue ball =120/520=3/13 White ball= 300/520= 15/26	2		
VII	From the formula for compound interest, we know, C.I = $P(1+R/100)^t - P$	1	7	7
	Here, P = 50,000 ; R = 10% ; T = 5 years ; C.I=? So, Compound Interest will be-	2		
	C.I. = $50000(1+10/100)^5 - 50000 = 80525.50 - 50000$	2		
	So, compound interest will be Rs. 30525.50 while the total accumulated wealth (A) will be Rs. 80525.50 at the end of 5 years with an interest rate of 10% per annum.	2		

<table>
<tr>
<td>VIII</td>
<td>
<table>
<tr><th colspan="3">Simple Interest and Compound Interest Differences</th></tr>
<tr><th>Parameter</th><th>Simple Interest</th><th>Compound Interest</th></tr>
<tr><td>Definition</td><td>Simple Interest can be defined as the sum paid back for using the borrowed money, over a fixed period of time.</td><td>Compound Interest can be defi when the sum principal amo exceeds the due date for paymen with the rate of interest, for a pe time.</td></tr>
<tr><td>Formula</td><td>S.I. = $(P \times T \times R)/100$</td><td>C.I. = $P(1+R/100)^{t} - P$</td></tr>
<tr><td>Return Amount</td><td>The return is much lesser when compared to Compound Interest.</td><td>The return is much highe</td></tr>
<tr><td>Principal Amount</td><td>The principal amount is constant</td><td>The principal amount keeps on during the entire borrowing p</td></tr>
<tr><td>Growth</td><td>The growth remains quite uniform in this method.</td><td>The growth increases quite rap this method.</td></tr>
<tr><td>Interest Charged</td><td>The interest charged on is for the principal amount.</td><td>The interest charged on it is f principal and accumulated int</td></tr>
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</td>
<td>1*7</td>
<td>7</td>
<td>7</td>
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IX	Let us consider 1 kg of grocery bag. Its actual weight is 85% of 1000 gm = 850 gm. Let the cost price of each gram be Re. 1. Then the CP of each bag = Rs. 850. SP of 1 kg of bag = 120% of the true CP Therefore, SP = 120/100 * 1000 = Rs. 1200 Gain = 1200 – 850 = 350 Hence Gain % = 350/850 * 100 = 41.17%	1 3 3	7	7
X	Total number of fruits shopkeeper bought = 1600 + 1400 = 3000 Number of rotten oranges = 15% of 1600 = 15/100 × 1600 = 24000/100 = 240 Number of rotten bananas = 8% of 1400 = 8/100 × 1400 = 11200/100 = 112 Therefore, total number of rotten fruits = 240 + 112 = 352 Therefore Number of fruits in good condition = 3000 - 352 = 2648 Therefore Percentage of fruits in good condition = (2648/3000 × 100)% = (264800/3000)% = 88.2%	2 2 3	7	7
XI	$\square = \square\square\square\square^3\square, \quad \square = \square\square\square\square^3\square$ $\frac{\square\square}{\square\square} = \frac{\square(\square\square\square\square^3\square)}{\square\square} = 3\square\square\square\square^2\square(-\square\square\square\square)$	2.5	7	7

	$\frac{\square\square}{\square\square} = \frac{\square(\square\square\square\square^{3}\square)}{\square\square} = 3\square\square\square\square^{2}\square(\square\square\square\square)$	2.5		
	$\frac{\square\square}{\square\square} = \frac{\square\square}{\square\square}\frac{\square\square}{\square\square} = \frac{3\square\square\square\square^{2}\square(\square\square\square\square)}{3\square\square\square\square^{2}\square(-\square\square\square\square)} = -\square\square\square\square$	2		
XII	$\frac{\square}{\square\square}(e^{\square\square\square\square-\square\square\square\square})$	2	7	7
	Let $\square = \square\square\square\square - \square\square\square\square$, $\frac{\square\square}{\square\square} = \square\square\square\square + \square\square\square\square$	3		
	$\frac{\square}{\square\square}(e^{u}) = e^{u}(\square\square\square\square + \square\square\square\square)$ $= e^{\square\square\square\square-\square\square\square\square}(\square\square\square\square + \square\square\square\square)$	2		
XIII	$\int (\log x)^2 \,\square\, \square\square$	1	7	7
	$= (\square\square\square\square)^2 \frac{\square^2}{2} - 2\int \frac{\square\square\square\square.\square^2}{2\square}\square\square$ $= \frac{\square^2}{2}(\square\square\square\square)^2 - \int \square\square\square\square.\square\square\square = \frac{\square^2}{2}(\square\square\square\square)^2$	2		
	$-\square\square\square\square.\frac{\square^2}{2} + \int \frac{1}{\square}\frac{\square^2}{2}\square\square$	2		
	$= \frac{\square^2}{2}(\square\square\square\square)^2 - \square\square\square\square.\frac{\square^2}{2} + \frac{\square^2}{4} + \square$ $= \frac{\square^2}{2}\left[(\square\square\square\square)^2 - \square\square\square\square. + \frac{1}{2}\right] + \square$	2		
XIV	$\int \square^2\square^{2\square}\square\square = \square^2\int \square^{2\square}\square\square - \int\left(\frac{\square(\square^2)}{\square\square}\int \square^{2\square}\square\square\right)\square\square$	2	7	7

	$= \square^{2}\dfrac{\square^{2\square}}{2} - \int\left(2\square\dfrac{\square^{2\square}}{2}\right)\square\square = \square^{2}\dfrac{\square^{2\square}}{2} - \int \square\square^{2\square}\square\square$	3		
	$= \dfrac{\square^{2\square}\square^{2}}{2} - \square\dfrac{\square^{2\square}}{2} - \dfrac{1}{2}\int \square^{2\square}\square\square$ $= \dfrac{\square^{2\square}\square^{2}}{2} - \square\dfrac{\square^{2\square}}{2} - \dfrac{\square^{2\square}}{4}$	2		